

UNIT-I

Classification

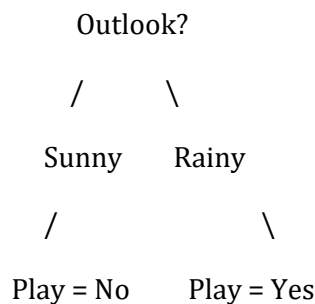
Classification is a supervised learning technique used to assign data items into predefined classes or categories.

It requires a labeled dataset where each instance belongs to a known class.

The goal is to learn a model that can predict the class of unseen data.

Common algorithms include Decision Trees, Naive Bayes, K-Nearest Neighbors, and Support Vector Machines.

1. Decision Tree Diagram



- Root Node → Attribute (Outlook)
- Branch → Condition (Sunny/Rainy)
- Leaf → Class (Yes/No)

Steps involved: Data collection, preprocessing, model training, testing, and evaluation.

Evaluation metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix.

Applications: Email spam detection, disease diagnosis, credit risk analysis.

Advantages: Easy interpretation (especially decision trees), high accuracy for structured data.

Challenges: Overfitting, handling imbalanced datasets, feature selection.

Clustering

Clustering is an unsupervised learning technique used to group similar data points together.

Unlike classification, clustering does not require labeled data.

The objective is to maximize similarity within clusters and minimize similarity between clusters.

Types of clustering: Partitioning (K-means), Hierarchical, Density-based (DBSCAN), Grid-based.

K-means is the most widely used algorithm and works by minimizing intra-cluster variance.

Hierarchical clustering builds a tree-like structure called dendrogram.

Evaluation: Silhouette score, Davies-Bouldin index.

Applications: Customer segmentation, image segmentation, document grouping.

Challenges: Choosing optimal number of clusters, handling noise and outliers.

2. Clustering Diagram

Cluster 1: ● ● ● (close together)

Cluster 2: ● ● ● (separate group far away)

Label:

- Cluster 1 (Group A)
- Cluster 2 (Group B)

Association Rule Mining

Association rule mining discovers interesting relationships among variables in large datasets.

It is widely used in market basket analysis.

Rules are of the form $X \rightarrow Y$, where X and Y are itemsets.

Key metrics: Support (frequency), Confidence (strength), Lift (correlation).

Apriori algorithm is commonly used for finding frequent itemsets.

FP-Growth is an efficient alternative that avoids candidate generation.

Steps: Generate frequent itemsets → Generate strong association rules.

Applications: Retail, recommendation systems, web usage mining.

Challenges: Large number of rules, selecting meaningful patterns.

Bread → Butter ; Milk → Bread

Regression

Regression is a supervised learning technique used to predict continuous values.

It models the relationship between dependent and independent variables.

Types: Linear Regression, Multiple Regression, Polynomial Regression.

Linear regression assumes a linear relationship between variables.

Model evaluation: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared.

Applications: Sales forecasting, price prediction, demand analysis.

Advantages: Simple and interpretable.

Challenges: Assumption violations, multicollinearity, overfitting

Points scattered → Line passing through them

Outlier Detection

Outlier detection identifies data points that significantly differ from the rest.

Outliers may indicate errors, fraud, or rare events.

Types: Global outliers, contextual outliers, collective outliers.

Techniques: Statistical methods, distance-based methods, density-based methods.

Z-score and IQR are common statistical approaches.

Applications: Fraud detection, network security, fault detection.

Cluster of points + 1 point far away

● ● ● ● ● ● (Outlier)

Challenges: Defining thresholds, distinguishing noise from useful anomalies.

Handling outliers: Removal, transformation, or separate analysis.

Data Preprocessing

Definition:

Data preprocessing is the process of preparing raw data for analysis by improving its quality, consistency, and usefulness. It is a crucial step in data mining to ensure accurate and efficient results.

1. Data Cleaning

Purpose: Handle missing, noisy, and inconsistent data

Key Activities:

- **Handling Missing Values:**
 - Ignore tuples
 - Fill with mean/median/mode
 - Use prediction (e.g., regression)
- **Noise Removal:**
 - Smoothing techniques (binning, clustering)
- **Outlier Detection:**
 - Identify abnormal data points
- **Inconsistency Correction:**
 - Resolve conflicting data (e.g., different formats) Date

Example:

Missing student marks → replaced with average marks

2. Data Integration

Purpose: Combine data from multiple sources into a unified dataset

Key Activities:

- Merging databases, files, or data warehouses
- Schema integration (matching attributes)
- Entity identification (same object from different sources)

- Removing redundancy and duplicates

Challenges:

- Data conflicts (different formats, units)
- Duplicate records

Example:

Combining student data from admission system + exam system

3. Data Transformation

Purpose: Convert data into suitable format for mining

Key Techniques:

- **Normalization:**
 - Scale values to a small range (0–1)
- **Aggregation:**
 - Summarization (daily → monthly data)
- **Generalization:**
 - Replace low-level data with higher-level concepts
- **Encoding:**
 - Convert categorical → numerical (e.g., Male=0, Female=1)

Example:

Marks scaled from 0–100 → 0–1 range

4. Data Reduction

Purpose: Reduce data size while maintaining integrity

Key Techniques:

- **Dimensionality Reduction:**
 - Remove irrelevant features
- **Numerosity Reduction:**
 - Use models instead of full data
- **Data Compression:**
 - Reduce storage size
- **Sampling:**
 - Use subset of data

Benefits:

- Faster processing
- Reduced storage
- Improved model performance

Example:

Selecting only important attributes like marks and attendance

Conclusion

Data preprocessing improves **data quality, efficiency, and accuracy** of data mining. Without preprocessing, results may be unreliable and misleading.

Concept of Datasets and Attributes**Definition**

A **dataset** is a collection of data objects (records), where each object is described by a set of **attributes** (features). It is the basic input for data mining and machine learning tasks.

1. Dataset (Data Collection)**Definition:**

A dataset is a structured collection of data, typically represented in tabular form.

Structure:

- **Rows (Tuples / Records):** Individual data objects
- **Columns (Attributes / Features):** Properties describing each object

Example:**Student_ID Name Age Marks Result**

101	Ravi	20	85	Pass
102	Meena	19	45	Fail

Types of Datasets

- **Structured Data:**
 - Organized in rows and columns (e.g., databases)
 - **Semi-Structured Data:**
 - Flexible format (e.g., XML, JSON)
 - **Unstructured Data:**
 - No fixed format (e.g., text, images, videos)
-

2. Attributes (Features / Variables)

Definition:

Attributes are the properties or characteristics used to describe each data object.

Types of Attributes

1. Nominal (Categorical)

- No order or ranking
- Example: Gender (Male/Female), Color

2. Ordinal

- Ordered categories
- Example: Rank (1st, 2nd), Grade (A, B, C)

3. Numeric

- Quantitative values
 - Two types:
 - **Interval:** No true zero (Temperature)
 - **Ratio:** Has true zero (Age, Salary)
-

3. Attribute Roles in Data Mining

- **Input Attributes (Independent variables):**
 - Used to predict output
- **Output Attribute (Target/Class):**
 - Variable to be predicted

- **Identifier Attribute:**
 - Unique ID (not used in analysis)

Example:

- Input: Age, Marks
 - Output: Result (Pass/Fail)
-

4. Importance in Data Mining

- Determines the **quality of analysis**
 - Helps in **model building and prediction**
 - Influences **accuracy and performance**
 - Proper selection of attributes reduces complexity
-

Conclusion

Datasets and attributes form the **foundation of data mining**. Understanding their structure and types is essential for effective data preprocessing, modeling, and analysis.

Supporting Techniques in Data Mining

Data mining relies on several core disciplines to extract meaningful patterns from data. The three major supporting techniques are **Statistics, Machine Learning, and Databases**.

1. Statistics

- Provides the **mathematical foundation** for data analysis
 - Helps in **data summarization, probability estimation, and hypothesis testing**
 - Used to identify patterns, trends, and relationships
 - Supports decision-making through measures like **mean, variance, correlation**
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2. Machine Learning

- Enables systems to **learn patterns automatically from data**
 - Provides algorithms for tasks like **classification, clustering, and prediction**
 - Improves performance through **training and experience**
 - Examples: Decision Trees, Neural Networks, Support Vector Machines
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3. Databases

- Responsible for **data storage, retrieval, and management**
 - Provides efficient handling of **large-scale data**
 - Uses technologies like **Data Warehouses and DBMS**
 - Supports preprocessing tasks such as **querying, indexing, and integration**
-

Conclusion

These techniques work together to ensure **accurate, efficient, and scalable data mining processes**.

Components of Data Mining Algorithms

Data mining algorithms are designed using three key components: **Model, Preference, and Search**. These components work together to discover meaningful patterns from data.

1. Model

- Defines the **type of pattern or structure** to be discovered
- Represents how data relationships are expressed
- Different models are used for different tasks:
 - **Classification Model** → Decision Trees, SVM
 - **Clustering Model** → K-means
 - **Association Model** → Rule-based patterns

Key Point: Model determines *what kind of knowledge* is extracted.

2. Preference (Evaluation Criteria)

- Specifies **which patterns are interesting or useful**
- Uses measures to evaluate model quality:
 - Accuracy (for classification)
 - Support & Confidence (for association rules)
 - Error rate or cost functions

Key Point: Preference determines *which model is best*.

3. Search

- Refers to the **process of exploring possible models**
- Since many possible patterns exist, efficient search is required
- Types of search strategies:
 - Exhaustive search (checks all possibilities)
 - Heuristic search (faster, approximate)
 - Greedy methods (step-by-step optimization)

Key Point: Search determines *how models are found efficiently*.

Conclusion

The effectiveness of a data mining algorithm depends on the proper design of:

- **Model (representation)**
- **Preference (evaluation)**
- **Search (optimization)**

Together, they ensure accurate and efficient pattern discovery.

Real-Time Example: Online Shopping Recommendation System

Scenario:

An e-commerce platform (like Amazon/Flipkart) wants to **recommend products to users** based on their behavior.

1. Model (What to learn?)

- The system uses a **Recommendation Model** (e.g., association rules or collaborative filtering)

Example:

- “Users who buy *mobile* also buy *earphones*”

□ Model Representation:

- If {Mobile} → {Earphones}

Meaning:

- The model captures the **relationship between products**
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2. Preference (Which model is best?)

- The system evaluates rules using **interestingness measures**

Example:

- Support = 30% (30% users bought both items)
- Confidence = 80% (80% who bought mobile also bought earphones)

□ The system selects rules with:

- High confidence
- High support

Meaning:

- Only **strong and useful patterns** are selected

3. Search (How to find the model?)

- The system searches through **millions of product combinations**

Example:

- Uses **Apriori algorithm** or **FP-Growth**
- Instead of checking all combinations:
 - Eliminates low-frequency itemsets early (pruning)

□ Efficient Search:

- Finds best rules without checking all possibilities
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Final Outcome

- When a user adds a mobile to cart:
 - System recommends → *"Frequently bought together: Earphones"*

Component	Role	Example
Model	Type of pattern	Mobile → Earphones
Preference	Evaluate quality	High confidence rule
Search	Find patterns	Apriori algorithm

Real-Time Example: Student Performance Prediction

Scenario:

A college wants to **predict whether a student will Pass/Fail** based on internal performance and behavior.

1. Model (What to learn?)

- The system uses a **Classification Model** (e.g., Decision Tree / Naive Bayes)

Input Attributes:

- Attendance
- Internal Marks
- Assignment Submission
- Study Hours

Output (Target):

- Result → Pass / Fail

Example Rule (Decision Tree):

- If Attendance > 75% AND Marks > 50 → Pass
- Else → Fail

☐ Meaning:

- Model defines the **relationship between student data and result**
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2. Preference (Which model is best?)

- The system evaluates models using **accuracy and error rate**

Example:

- Model A Accuracy = 85%
- Model B Accuracy = 92%

☐ Select Model B (higher accuracy)

Other Measures:

- Precision (correct pass predictions)
- Recall (detecting actual failures)

☐ Meaning:

- Preference ensures we choose the **most reliable prediction model**
-

3. Search (How to find the model?)

- The system explores multiple possible models

Example:

- Decision Tree tries different splits:
 - Split by Attendance first
 - Split by Marks first
- Uses **greedy search (Information Gain)** to select best attribute

□ Instead of checking all combinations:

- Chooses best split step-by-step

□ **Meaning:**

- Search helps **efficiently build the best model**
-

Final Outcome

- System predicts:
 - Student A → Pass
 - Student B → At Risk (Fail)

□ Faculty can:

- Identify weak students early
- Provide remedial classes

Attendance,InternalMarks,Assignments,StudyHours,Result

85,78,Yes,3,Pass

60,45,No,1,Fail

90,88,Yes,4,Pass

70,55,Yes,2,Pass

50,35,No,1,Fail

95,92,Yes,5,Pass

65,48,No,2,Fail

80,67,Yes,3,Pass

55,40,No,1,Fail

75,60,Yes,2,Pass

WEKA Implementation (Step-by-Step for Lab)

Step 1: Prepare Dataset

- Open Excel → Save as **CSV**
 - Convert CSV → ARFF (or open CSV directly in WEKA)
-

Step 2: Open WEKA

- Open **WEKA Explorer**
 - Click **Open file** → select dataset
-

Step 3: Preprocess

- Ensure:
 - Attributes are correctly detected
 - Result is **Nominal (Pass/Fail)**
-

Step 4: Apply Algorithm

- Go to **Classify tab**
 - Choose:
 - `trees` → J48 (Decision Tree)
-

Step 5: Train Model

- Select:
 - **Use training set OR**
 - **10-fold cross-validation**
 - Click **Start**
-

Step 6: Output

You will get:

- Accuracy %
- Confusion Matrix
- Decision Tree structure

Expected Decision Rule

IF Attendance > 70 AND InternalMarks > 50 → Pass

ELSE → Fail

The rule is derived using a decision tree algorithm, which selects Attendance and Internal Marks as the most important attributes based on information gain, and forms the rule by analyzing how data is split to maximize classification accuracy.”

🔍 Step 1: Dataset Summary

We use this target:

- **Result → Pass / Fail**

From dataset (10 records):

- Pass = 6
 - Fail = 4
-

🔍 Step 2: Calculate Entropy of Dataset

🔍 Formula:

$$\text{Entropy}(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

Where:

- $p_1 = \text{Pass} = 6/10 = 0.6$
- $p_2 = \text{Fail} = 4/10 = 0.4$

🔍 Calculation:

$$\text{Entropy}(S) = -(0.6 \log_2 0.6 + 0.4 \log_2 0.4)$$

$$\log_2(0.6) \approx -0.737$$

$$\log_2(0.4) \approx -1.322$$

$$\begin{aligned}\text{Entropy}(S) &= -(0.6 \times -0.737 + 0.4 \times -1.322) \\ &= 0.442 + 0.529 \\ &= 0.971\end{aligned}$$

$$\square \text{ Entropy}(S) = 0.971$$

🔍 Step 3: Choose Attribute (Attendance)

Split data into:

- **Attendance > 70**
- **Attendance ≤ 70**

🔍 Subset 1: Attendance > 70

Records:

- Total = 5
- Pass = 5
- Fail = 0

$$\text{Entropy} = -(1 \log_2 1 + 0 \log_2 0) = 0$$

$$\square \text{ Entropy} = 0 \text{ (Pure node)}$$

🔍 Subset 2: Attendance ≤ 70

Records:

- Total = 5
- Pass = 1
- Fail = 4

$$p_1 = 1/5 = 0.2$$

$$p_2 = 4/5 = 0.8$$

$$\text{Entropy} = -(0.2 \log_2 0.2 + 0.8 \log_2 0.8)$$

$$\log_2(0.2) \approx -2.322$$

$$\log_2(0.8) \approx -0.322$$

$$\begin{aligned}\text{Entropy} &= (0.2 \times 2.322 + 0.8 \times 0.322) \\ &= 0.464 + 0.258 \\ &= 0.722\end{aligned}$$

□ **Entropy = 0.722**

🔍 Step 4: Compute Weighted Entropy

$$\begin{aligned}\text{Weighted Entropy} &= (5/10 \times 0) + (5/10 \times 0.722) \\ &= 0 + 0.361 \\ &= 0.361\end{aligned}$$

🔍 Step 5: Information Gain

🔍 **Formula:**

$$\text{IG}(S, \text{Attendance}) = \text{Entropy}(S) - \text{Weighted Entropy}$$

🔍 **Calculation:**

$$\begin{aligned}\text{IG} &= 0.971 - 0.361 \\ &= 0.610\end{aligned}$$

□ **Information Gain (Attendance) = 0.610**

🔍 Step 6: Interpretation

- High IG (0.610) → **Good split**
 - Attendance strongly separates Pass/Fail
 - Hence chosen as **root node**
-

🔍 Step 7: Final Conclusion (Exam Point)

□ *“Attendance has the highest Information Gain (0.610), so it is selected as the root node in the decision tree. It creates a near-pure split, making it the most effective attribute for classification.”*

Precision, Recall, and F1-Score Notes

1. Introduction

In classification problems, evaluating model performance is very important. Accuracy alone is not sufficient, especially when the dataset is imbalanced. Therefore, we use additional metrics such as **Precision, Recall, and F1-score**.

2. Confusion Matrix

A confusion matrix is used to evaluate classification results.

		Predicted Pass	Predicted Fail
Actual	Pass	TP	FN
	Fail	FP	TN

- **TP (True Positive):** Correctly predicted Pass
- **TN (True Negative):** Correctly predicted Fail
- **FP (False Positive):** Incorrectly predicted Pass
- **FN (False Negative):** Incorrectly predicted Fail

3. Precision

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Example:

$$\text{Precision} = 5 / (5 + 1) = 5/6 = 0.83$$

Interpretation:

- Precision measures how accurate the positive predictions are.
- Out of all predicted Pass students, **83% are actually Pass.**

4. Recall

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Example:

$$\text{Recall} = 5 / (5 + 1) = 5/6 = 0.83$$

Interpretation:

- Recall measures how well the model identifies actual positives.
- Out of all actual Pass students, **83% are correctly identified.**

5. F1-Score

Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Example:

$$\text{F1} = 2 \times (0.83 \times 0.83) / (0.83 + 0.83)$$

$$\text{F1} = 0.83$$

Interpretation:

- F1-score is the **harmonic mean** of Precision and Recall.
 - It provides a balance between both metrics.
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6. Summary Table

Metric	Value
Precision	83%
Recall	83%
F1-Score	83%

7. Key Points

- Precision focuses on **prediction correctness**
 - Recall focuses on **capturing all actual positives**
 - F1-score balances both
 - Important in **real-world applications like student performance prediction**
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8. Applications

- Identifying weak students
 - Medical diagnosis
 - Fraud detection
 - Spam email filtering
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9. Conclusion

Precision, Recall, and F1-score provide a better understanding of model performance compared to accuracy. They are essential when dealing with **imbalanced datasets**.